ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 5 · Benchmarks and Forecast Accuracy

Professor: Angel Mora Segura

30 September 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA ✓
3	23 Sep	Lags, ACF, PACF ✓
4	24 Sep	Decomposition, STL, Box-Cox · Checkpoint 1 ✓
5	30 Sep	Benchmarks: Naive & Seasonal Naive **TODAY**
6	01 Oct	Simple exponential smoothing
7	07 Oct	Holt, damped trend, Holt-Winters
8	08 Oct	ETS taxonomy + model selection
9	15 Oct	**Midterm (intermediate test) — 20% · Checkpoint 2**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier
17	25 Nov	Integrated Forecasting Workshop · Checkpoint 3
18	02 Dec	**Group presentations — 30% · double session, part 1**
19	02 Dec	**Group presentations — 30% · double session, part 2**
20	11 Dec	**Final exam in R — 40%**

TODAY'S SESSION · BLOCK 2 BEGINS

Setting the Bar

Four simple forecasting methods, an honest way to test them, and the accuracy measures that decide whether anything more complicated is worth it.

FPP3 · Chapter 5

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Produce forecasts with the **mean, naive, seasonal naive and drift** methods, and say which is the fair benchmark for a given series.

2. Split a series into **training and test** periods correctly, and explain why accuracy on the training data is not evidence.

3. Check **residuals** for the four properties a good forecasting method leaves behind, including the Ljung-Box test.

4. Compute and interpret **MAE, RMSE, MAPE and MASE**, and choose the one your business decision actually cares about.

5. **Fix the benchmark for your case** — the method and the number that every later model is judged against.

BLOCK 1 · 30 MINUTES

Concept + R Application

Forecast first, then learn how to be honest about how good the forecast is.

IN CLASS: THE CORE IDEAS · ANNEX + QUARTO: EQUATIONS, R DETAILS AND THE REST

Why Start With the Simplest Methods?

A benchmark is a method so simple that nobody would pay for it. Its job is not to be good — it is to tell you whether anything else is. In forecasting competitions, simple methods regularly beat sophisticated ones.

01

It is the honest zero

A model that cannot beat “same as last year” has added cost and complexity, and nothing else. You only find that out if you ran the comparison.

02

It is often what the client already does

Many businesses plan with last year’s figures or a running average. The benchmark is frequently their current method — the thing your recommendation must improve on.

03

It is sometimes the answer

For a noisy, short or white-noise series, a benchmark can be the best available method. Recommending it, with evidence, is a legitimate result.

Rule for the rest of the course: **no model is presented without its benchmark next to it.**

Four Benchmark Methods

Each makes one simple assumption about the future. Choosing between them is the first use of everything you learned in Block 1.

Mean

`MEAN(y)`

Every future value equals the average of the history.

Fair benchmark for: A series with no trend and no seasonality — rare in business.

Naive

`NAIVE(y)`

Every future value equals the last observation.

Fair benchmark for: Random walks: prices, exchange rates, anything where the latest value carries all the information.

Seasonal naive

`SNAIVE(y)`

Each future value equals the value from the same season last year.

Fair benchmark for: Any strongly seasonal series — the default benchmark for most demand data.

Drift

`RW(y ~ drift())`

The last value, extended along the average historical change.

Fair benchmark for: A series with a steady trend and no seasonality.

Pick the benchmark from what Block 1 told you: seasonal series → seasonal naive; trend only → drift; neither → mean or naive.

EXAMPLE HAPPY COW · TRAIN ON 2024–2025, FORECAST 2026

The Four Methods on the Same Series



Only one of them looks like the series.

- **Mean** and **naive** are flat lines: they ignore the summer peak entirely.
- **Drift** extends the average change — nearly flat here, because the series starts and ends in winter.
- **Seasonal naive** reproduces the shape, and misses only because 2026 grew a little above 2025.
- That residual gap — the growth — is exactly what a better model will have to capture to earn its place.

R APPLICATION

The Workflow You Will Use for Every Model

```
train <- sales |>
  filter_index(. ~ "2025 Dec")
fit <- train |>
  model(
    mean   = MEAN(units),
    naive  = NAIVE(units),
    snaive = SNAIVE(units),
    drift  = RW(units ~ drift())
  )
fc <- fit |> forecast(h = 12)
fc |> autoplot(sales, level = NULL)
```

Three verbs, always the same

- `model()` fits one or more methods at once. The result is a *mable*: one row per series, one column per model.
- `forecast()` produces a *fable*: a distribution for every future period, not just a number.
- `accuracy()` compares forecasts against the held-out actuals.
- From Session 6 onwards only the lines inside `model()` change. ETS, ARIMA and the rest slot into exactly the same pipeline.

Transformations carry through. Write `SNAIVE(log(units))` and the forecast is back-transformed automatically, with the bias adjustment applied.

`model()` → `forecast()` → `accuracy()`: the same three verbs for every model from here to Session 20.

Training and Test Sets

A model that fits the past well is not therefore a good forecaster. The only fair test is data the model has never seen: hold back the most recent stretch, fit on the rest, and compare the forecasts with what actually happened.



Training · 24 months · fit here **Test · 12 months**

RULE 1

Split by time, never at random

The test set is always the **end** of the series. A random split lets the model see the future, and makes every method look better than it is.

RULE 2

Make the test as long as the horizon

If the business needs 12 months ahead, test on at least 12. A test shorter than the horizon measures the wrong thing.

RULE 3

Touch the test set once

Choosing a model because it did well on the test set turns the test into training. Decide on the training data; confirm on the test.

What Good Residuals Look Like

Two properties are essential; two are convenient. A method that fails the first two can be improved. A method that fails only the last two is fine for point forecasts, but its prediction intervals need care.

ESSENTIAL

Uncorrelated

If residuals are autocorrelated, information is left in them that should be in the forecast. Check the ACF of the residuals.

ESSENTIAL

Zero mean

A non-zero mean means the forecasts are biased. The fix is simple: add the mean back.

CONVENIENT

Constant variance

Makes prediction intervals reliable. If the spread grows with the level, a transformation usually solves it.

CONVENIENT

Normally distributed

Makes the usual prediction intervals valid. If not, use bootstrapped intervals instead.

```
fit |> select(snaive) |> gg_tsresiduals()
```

One line draws the time plot, the ACF and the histogram of the residuals together — all four checks at once.

Measuring Forecast Accuracy

A forecast error is the actual test value minus the forecast. The measures differ in how they summarise those errors — and each one answers a slightly different business question.

MAE Mean absolute error

The average size of the miss, in the units of the data. Easy to explain: “on average we are 380 units out”.

Scale-dependent: cannot compare series of different sizes.

RMSE Root mean squared error

Squares the errors first, so one large miss counts far more than several small ones.

Use it when a big error is disproportionately expensive.

MAPE Mean absolute percentage error

Errors as a percentage of the actual value. Intuitive for managers.

Breaks near zero and penalises over- and under-forecasts unequally. Avoid for intermittent demand.

MASE Mean absolute scaled error

MAE divided by the in-sample error of a seasonal naive forecast. Below 1 means better than that naive in-sample benchmark.

Scale-free and safe near zero: the default for comparing across series.

EXAMPLE THE FOUR BENCHMARKS, SCORED ON THE 2026 TEST SET

Reading an Accuracy Table

Method	RMSE	MAE	MAPE	MASE
Seasonal naive	**411**	**386**	**9.3%**	**1.57**
Mean	1,628	1,343	32.1%	5.48
Drift	2,376	1,814	35.2%	7.40
Naive	2,447	1,896	37.3%	7.74

```
fc |> accuracy(sales) |>
  select(.model, RMSE, MAE, MAPE, MASE) |>
  arrange(MASE)
```

Seasonal naive wins on every measure, by a wide margin.

- Every measure agrees on the ranking here. When they disagree, choose the one that matches your cost of error — and say which in the log.
- Seasonal naive's MASE is 1.57, **above 1**: its 2026 errors are larger than its own in-sample errors, because 2026 grew. That is not failure — it is the room a better model can win.
- **This is now the bar.** Any model proposed for this series must beat an MAE of 386 units on 2026.

Time Series Cross-Validation

One train/test split gives one score, and it depends on which year happened to be held out. Cross-validation repeats the exercise many times: fit on an expanding window, forecast the next period, move the origin forward, and average the errors.




```
sales |>
  stretch_tsibble(.init = 24,
                  .step = 1) |>
  model(snaive = SNAIVE(units)) |>
  forecast(h = 1) |>
  accuracy(sales)
```

- More reliable than a single split, especially with short series like Clean Air's.
- Set `h` to your business horizon, not to 1, if that is what the decision needs.

BLOCK 2 · 30 MINUTES

Business Case Lab

Fit the benchmarks to your own series, test them honestly, and fix the number your project has to beat.

WHAT TO EXPECT IN EACH CASE

The Fair Benchmark for Each Case

CASE 1 · RETAIL

Happy Cow Ice Cream

Daily data with a weekly cycle: seasonal naive with m = 7 — "same day last week". Hold out the last four weeks to match the delivery horizon.

Watch for: with seven months of data, cross-validation gives a far more stable score than a single split.

CASE 2 · AIRLINES

Balearic Airlines

Seasonal naive with m = 7 is the natural benchmark. The operational horizon is about 9–15 days, so test on at least two weeks.

Watch for: the holidays. Seasonal naive copies last week's holiday into this week. Check the errors around them separately.

CASE 3 · MANUFACTURING

Clean Air Limited

Monthly, m = 12: seasonal naive is the benchmark. With 36 months, a 12-month test leaves only 24 to train on.

Watch for: the structural break. If it falls in your training data, seasonal naive may copy a COVID month forward. Decide how to handle it and record the choice.

CASE 4 · E-COMMERCE

Wayfair

Fit the benchmarks to the Decor category and its four SKUs at once — `model()` does this automatically across the key.

Watch for: MAPE is undefined for zero-demand months. Use MASE, and compare the sparse and dense SKUs separately.

BLOCK 3 · 20 MINUTES

Case Clinic

Three or four teams: your benchmark, its score, and why that measure.

SESSION 5 RECAP

What to Take Away

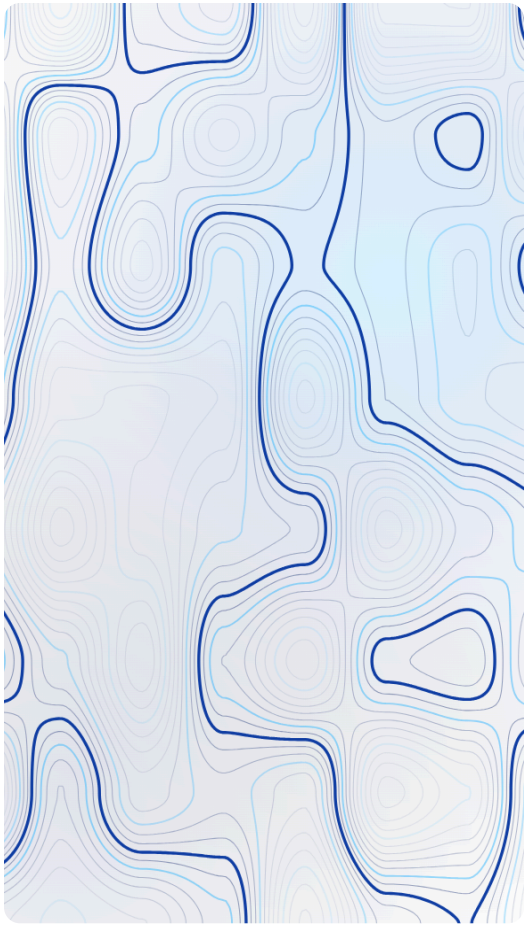
Benchmarks first, always. Mean, naive, seasonal naive and drift — the fair one depends on what Block 1 told you about the series.

Test on the future. Split by time, test for at least the business horizon, and use the test set once.

Residuals diagnose, accuracy decides. Uncorrelated with zero mean is essential; Ljung-Box checks it formally.

Choose the measure from the decision. MAE to explain, RMSE when big misses hurt, MASE to compare, MAPE with care.

Next session: simple exponential smoothing — a weighted average of the past, and the first model that has to beat your number.



END OF SESSION 05

TIME SERIES ANALYSIS

Benchmarks and Forecast Accuracy

NEXT · SESSION 6 · 01 OCTOBER

Simple exponential smoothing

The first real forecasting model — and the first one that has to beat today's benchmark to earn its place.

The annex after this slide holds the reference material, including the R cheat sheet on slide 31.

Angel Mora Segura · IE University

30 September 2026

ANNEX · REFERENCE MATERIAL

For Study, Not for Class

Equations, R details, the cheat sheet and the common mistakes. Everything here is also developed in the session’s Quarto document.

Forecasting With a Decomposition

Session 4 pays off here. Split the series with STL, forecast the seasonally adjusted part with a simple method, then add the seasonal component back — a transparent method that is often surprisingly hard to beat.

1. **Decompose**
   STL separates the seasonal component from the seasonally adjusted series.

2. **Forecast the adjusted part**
   Any non-seasonal method: naive, drift, or later ETS.

3. **Forecast the season**
   Seasonal naive on the seasonal component — last year's pattern, repeated.

4. **Recombine**
   Add (or multiply, on the original scale) the two forecasts.

```
fit_dcmp <- train |>
  model(stlf = decomposition_model(
    STL(log(units)),
    RW(season_adjust ~ drift())
  ))
fit_dcmp |> forecast(h = 12) |>
  autoplot(sales)
```

- The seasonal component is handled automatically with `SNAIVE`; you only choose the method for the adjusted series.
- It captures what seasonal naive misses: **the growth**. Add it to your benchmark comparison as a fifth candidate.

Fitted Values and Residuals

A **fitted value** is the model's one-step forecast of an observation it has already seen. The **residual** is the difference between the observation and that fitted value — what the model failed to explain.

$$e_t = y_t - \hat{y}_{t|t-1}$$

For seasonal naive, the fitted value is the same month last year, so the residual is simply this year minus last year.

```
fit |> augment()
# .fitted  the one-step fitted value
# .resid   y minus .fitted
# .innov   residual on the transformed
#          scale - use this one
```

Why residuals matter

- If a pattern remains in the residuals, the model has left information on the table — and a better model could use it.
- Residuals are computed on the **training** data. They diagnose the model; they do not measure forecast accuracy.
- With a transformation, check the **innovation** residuals (`.innov`), which live on the transformed scale where the model was fitted.

Session 3's white noise returns: good residuals look like it. Now you know why we spent a session on the ACF.

A Formal Test: Ljung-Box

Looking at 24 ACF bars and expecting one or two to cross the band by chance is hard to do consistently. The Ljung-Box test combines the first *l* autocorrelations into a single statistic and asks whether, together, they are larger than white noise would produce.

```
fit |> augment() |>
  features(.innov, ljung_box,
           lag = 24, dof = 0)
#> .model   lb_stat  lb_pvalue
#> snaive      18.4      0.78
```

Choosing the lag. A common rule: *l* = 10 for non-seasonal data, *l* = 2*m* for seasonal data — 24 for monthly. `dof` is the number of estimated parameters: 0 for the benchmarks.

Reading the result

- **Large p-value** (above 0.05): no evidence the residuals differ from white noise. The model has used the structure it can see.
- **Small p-value**: the residuals are autocorrelated as a group. Something systematic is left — look at the ACF to find which lags.
- A passed test does not make a model good; it means the model is not *obviously* incomplete. Accuracy on the test set is still what decides.

Prediction Intervals From the Benchmarks

Every `forecast()` is a distribution, so every benchmark already comes with intervals. They are the first honest statement of how uncertain your project's forecast is.

WIDTH

Grows with the horizon

For naive methods the interval widens with every step. For seasonal naive it jumps once per season, because each year relies on a single past value.

ASSUMPTION

Built from the residuals

Intervals assume uncorrelated, normally distributed residuals. If the diagnostics fail, use `bootstrap = TRUE` in `forecast()`.

FOR YOUR CASE

The quantile you plan to

BalAir and Clean Air decide on a quantile, not a mean. `hilo(fc, 80)` gives you the interval your decision is built on.

Report the interval alongside every point forecast from now on. A number without its uncertainty is not a forecast.

EVERYTHING FROM TODAY, IN ONE PLACE

Session 5 R Cheat Sheet

```
# 1. split by time
train <- df |> filter_index(. ~ "2025 Dec")
# 2. fit the benchmarks
fit <- train |> model(
  mean   = MEAN(y),
  naive  = NAIVE(y),
  snaive = SNAIVE(y),
  drift  = RW(y ~ drift()),
  stlf   = decomposition_model(
    STL(log(y)), RW(season_adjust ~ drift())))
```

```
# 3. forecast and plot
fc <- fit |> forecast(h = 12)
fc |> autoplot(df, level = 80)
# 4. residual diagnostics
fit |> select(snaive) |> gg_tsresiduals()
fit |> augment() |>
  features(.innov, ljung_box, lag = 24)
# 5. accuracy on the test set
fc |> accuracy(df) |> arrange(MASE)
# 6. cross-validation
df |> stretch_tsibble(.init = 24) |>
  model(SNAIVE(y)) |> forecast(h = 1) |>
  accuracy(df)
```

Five Ways Accuracy Numbers Mislead

Almost every inflated accuracy claim in practice comes from one of these. They are worth checking before any number leaves your team.

Scoring on the training data	In-sample errors are always optimistic, because the model has already seen those observations. Only test-set or cross-validated errors count as evidence.
Choosing the model on the test set	Once the test set drives the choice, it is no longer a test. Select on training data or cross-validation; confirm once on the test.
A test shorter than the horizon	One month of test data says nothing about a 12-month decision.
Comparing MAE across different series	An error of 400 is small for one product and enormous for another. Use MASE to compare across series or teams.
Leaving the benchmark out	A model with a MAPE of 8% sounds good until seasonal naive scores 7%. Always show the benchmark in the same table.